

Discrete time Markov processes

Covered by these notes.

Discrete time Markov processes are stochastic processes which both have widespread applications and are relatively easy to simulate.

In this lecture we review some further such Markov processes, but first discuss briefly what a Markov process is.

These notes explain Markov processes with minimal mathematical rigor, please consult any textbook on Markov processes for more details.

Markov processes

Basic theory: Consider a stochastic process X_n .

The process is said to be **Markovian (or a Markov process)** if

$$\begin{aligned} P(X_{n+1} \in A | X_n = c, X_{n-1} = a_1, \dots, X_{n-q} = a_q) \\ = P(X_{n+1} \in A | X_n = c) \end{aligned}$$

for some subset of state space A .

I.e. the distribution of the state tomorrow does not depend on the state yesterday, as long as we know the state today.

Examples

$$X_{n+1} = aX_n + \varepsilon_{n+1}, \quad \varepsilon_{n+1} \sim \text{iid } N(0, 1),$$

is Markovian (as distribution of tomorrow depends only on today)

$$X_{n+1} = aX_n + bX_{n-1} + \varepsilon_{n+1}, \quad \varepsilon_{n+1} \sim \text{iid } N(0, 1),$$

is not Markovian (distribution of $X_{n+1}|X_n$ and $X_{n+1}|X_n, X_{n-1}$ are different).

A discrete time Markov process can be uniquely identified by its **transition distribution** $Q(X)$, so that $X_{n+1}|X_n \sim Q(X_n)$, e.g. in the former example above, $Q(X) = N(aX, 1)$.

If the transition distribution allows simulation, it is straight forward to simulate the Markov process:

1. Fix X_1 to some initial state and set $n \leftarrow 1$.
2. Simulate $X_{n+1} \sim Q(X_n)$.
3. Set $n \leftarrow n + 1$ and go back to step 2.

Stationarity

Some, but certainly not all, Markov processes exhibit a property which we shall refer to as **stationarity**.

Note: There exist several stationarity notions, here we consider a weak sense of stationarity.

Roughly speaking, a Markovian process X_n is said to be stationary if the distribution of $X_n|X_1 = a$, for all a , approaches some fixed probability distribution π (with finite variance) when time n goes to infinity. The distribution π is said to be the stationary distribution of process X_n .

Example: iid process

Let $X_n = \varepsilon_n$, $\varepsilon \sim \text{iid } N(0, \sigma^2)$ (i.e. $Q(X) = N(0, \sigma^2)$ (no dependence on X)). This process is stationary with stationary distribution $\pi = N(0, \sigma^2)$.

Example: Gaussian random walk (RW)

Let the process be defined by $X_1 = a$ and $X_{n+1} = X_n + \varepsilon_{n+1}$, $\varepsilon_{n+1} \sim \text{iid } N(0, \sigma^2)$.

Notice that

$$X_2 = a + \varepsilon_2$$

$$X_3 = a + \varepsilon_2 + \varepsilon_3$$

$$X_4 = a + \varepsilon_2 + \varepsilon_3 + \varepsilon_4$$

and generally for $n \geq 2$:

$$X_n = a + \sum_{i=2}^n \varepsilon_i.$$

From this representation and the fact that sums of Gaussian RVs is Gaussian, we have that

$$E(X_n | X_1 = a) = a,$$

$$\text{Var}(X_n | X_1 = a) = (n - 1)\sigma^2$$

and thus

$$X_n | X_1 = a \sim N(a, (n - 1)\sigma^2).$$

As $n \rightarrow \infty$, the variance of $X_n | X_1 = a$ goes to infinity, and the process is therefore not stationary.

Example: Gaussian AR(1) process

We consider a process very similar to the Gaussian RW, namely the Gaussian AR(1) process

$$X_{n+1} = \gamma + \phi X_n + \varepsilon_{n+1}, \quad \varepsilon_{n+1} \sim \text{iid } N(0, \sigma^2).$$

Let's look at a few iterations:

$$X_1 = a$$

$$X_2 = \gamma + \phi a + \varepsilon_2$$

$$X_3 = \gamma + \phi\gamma + \phi^2 a + \phi\varepsilon_2 + \varepsilon_3$$

and generally

$$X_n = a\phi^{n-1} + \gamma \sum_{i=0}^{n-2} \phi^i + \sum_{i=0}^{n-2} \phi^i \varepsilon_{n-i}$$

Clearly, $X_n|X_1 = a$ is a linear combination of Gaussian random variables, and hence also Gaussian. Thus it suffices to find the mean and variance:

$$E(X_n|X_1 = a) = a\phi^{n-1} + \gamma(1 - \phi^{n-1})/(1 - \phi).$$

$$\begin{aligned} \text{Var}(X_n|X_1 = a) &= \text{Var}(\sum_{i=0}^{n-2} \phi^i \varepsilon_{n-i}) = \\ \sigma^2 \sum_{i=0}^{n-2} \phi^{2i} &= \sigma^2(1 - (\phi^2)^{n-1})/(1 - \phi^2) \end{aligned}$$

Now we assume that $|\phi| < 1$, then $\lim_{n \rightarrow \infty} \phi^{n-1} = \lim_{n \rightarrow \infty} (\phi^2)^{n-1} = 0$, and

$$\lim_{n \rightarrow \infty} (X_n|X_1 = a) \sim N\left(\frac{\gamma}{1 - \phi}, \frac{\sigma^2}{1 - \phi^2}\right)$$

See R code: `AR1sim.R`

More on Stationarity

One reason we are interested in stationary Markov processes, is that we may approximate expectations of π by sample averages, i.e. we have a law of large numbers;

$$\frac{1}{T} \sum_{n=1}^T g(X_n) \rightarrow \int g(x) \pi(x) dx.$$

BUT in general:

$$Var \left(\frac{1}{T} \sum_{n=1}^T g(X_n) \right) \neq \frac{1}{T} Var(g(X_n))$$

as the X_n s are correlated.

In the last part of this course, we will look at algorithms for constructing Markov processes with an arbitrary stationary distribution π (e.g. the posterior distribution in Bayesian statistics or Boltzmann distribution in physics).

See R code: `AR1sim.R`